

Localization of single emitters with donut beams: a reproducible simulation study of MINFLUX-type patterns

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We present a reproducible simulation study of the localization of single fluorescent emitters with donut-shaped excitation beams in the triangular coordinate pattern (TCP) of MINFLUX: three donut zeros on a circle of diameter L plus a central exposure. Every computed number is produced by a script and was reproduced by an independent verifier. (i) At the TCP centre the Cramér–Rao bound (CRB) is discontinuous without background: the $r \rightarrow 0$ limit lies below the point value at $r = 0$ (Balzarotti Eq. S27), with ratio $2/\sqrt{5}$ for a quadratic zero; for $L = 50$ nm, $N = 100$ photons and a donut size parameter $\text{fwhm} = 300$ nm the limit is 1.605 nm against 1.802 nm; any finite signal-to-background ratio (SBR) removes the discontinuity. (ii) A high-numerical-aperture vectorial vortex donut of the correct handedness is reproduced by a scalar Laguerre–Gauss donut with matched zero curvature (CRB ratio 0.9962 at $L = 150$ nm), whereas the wrong handedness fills the zero to 0.8452 of the ring maximum. (iii) With SBR = 10 the maximum-likelihood estimator (MLE) is efficient at the centre ($L = 50$ nm, $N = 100$; $\sigma/\sigma_{\text{CRB}} = 0.991 \pm 0.005$); without background it is biased and superefficient. (iv) Iterative MINFLUX with 1000 photons and our protocol (L reduced from 150 to 25 nm, no background) reaches an error of 0.508 nm, 0.1606 times that of an ideal camera. (v) A finite zero depth ϵ sets an optimal $L \propto \text{fwhm}\sqrt{\epsilon}$, and a misaligned pattern analysed with the nominal model produces, at the centre, a mean bias of 0.747–0.783 times the displacement δ of the zeros ($\delta = 2$ –10 nm, $L = 100$ nm, SBR = 10), which an estimator using the true pattern removes.

I. INTRODUCTION

MINFLUX localizes a single emitter by probing it sequentially with an excitation intensity minimum placed at a few known positions and comparing the detected photon counts [1]. Because the fluorescence vanishes near the zero, the information per detected photon is large and the localization precision scales with the size L of the probing pattern rather than with the diffraction-limited width of a point-spread function; nanometre precision is reached with a few hundred photons, in two and three dimensions [1, 2]. The same probabilistic framework covers other sequential structured-illumination schemes [3, 4], multiphoton excitation with light minima [5] and nanoscale tracking [6]. The excitation minimum is usually a toroidal (donut) focus produced by a vortex phase mask; its zero depends on polarization and handedness [7, 8].

Theoretical analyses of MINFLUX, starting with the Supplementary Material of Ref. [1], use the scalar paraxial Laguerre–Gauss (LG) donut, a perfect zero, an exact knowledge of the pattern and the closed-form CRB at the pattern centre. Here we ask, with simulations whose every number is traceable to a script, how far these idealizations can be pushed. We address five questions:

R1 *Beam model*. How much does the scalar LG donut differ from the vectorial high-NA donut near the zero, and what does it change in the CRB? What happens with the wrong handedness or with linear polarization?

R2 *Cramér–Rao bound*. CRB maps of the TCP; the closed form at the centre versus the numerical computation; scaling with L , N and SBR; comparison with camera localization.

R3 *Estimators*. MLE versus least-mean-squares (LMS) estimators: bias and precision from Monte Carlo (MC), inside and outside the TCP, compared with the CRB.

R4 *Iterative MINFLUX*. Precision versus total photon budget when L is reduced over iterations, compared with a camera.

R5 *Non-idealities*. Finite zero depth, background, and misalignment of the pattern analysed with the nominal model.

The main results are: a precise statement of the discontinuity of the centre CRB and of the two possible definitions (Sec. IV, App. A); a quantitative equivalence between the vectorial donut and a curvature-matched LG donut, and the collapse of precision with a filled zero (Sec. III); the efficiency and the failure modes of the estimators (Sec. V); the photon scaling of iterative MINFLUX against an ideal camera (Secs. VI and VII); and the optimal L for a finite zero depth and the bias caused by misalignment (Sec. VIII). Section IX ranks these effects by their numerical impact, and Sec. X lists what remains open.

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II. MODEL AND CONVENTIONS

A. Beams

The scalar paraxial donut is the LG_{01} intensity in the parametrization of Balzarotti Eq. S17 [1], normalized to a ring peak of one:

$$I_{\text{LG}}(\rho) = 4e \ln 2 \frac{\rho^2}{\text{fwhm}^2} \exp\left(-4 \ln 2 \frac{\rho^2}{\text{fwhm}^2}\right) \quad (1)$$

$$\equiv e a \rho^2 e^{-a \rho^2}, \quad a = \frac{4 \ln 2}{\text{fwhm}^2}.$$

Here fwhm is a *size parameter*, not the full width at half maximum of any curve of the donut: the ring has a peak-to-peak diameter $\text{fwhm}/\sqrt{\ln 2}$, i.e. 360.3 nm for our default $\text{fwhm} = 300$ nm. Near the zero $I_{\text{LG}} \simeq e a \rho^2$ (quadratic).

The vectorial donut is computed with the Richards–Wolf (Debye) integral for an aplanatic objective [8]: a Gaussian beam with a $0-2\pi$ vortex phase $e^{i\phi}$ and circular polarization, focused with wavelength $\lambda = 640$ nm, numerical aperture $\text{NA} = 1.4$, refractive index $n = 1.518$ and filling factor $5/3$; the intensity is $|E_x|^2 + |E_y|^2 + |E_z|^2$. We call “correct” the handedness for which the circular polarization rotates in the same sense as the phase ramp [7, 8]; in our formulation, the product of the topological charge and the photon spin is $+1$. Figure 1(a,b) compares the profiles.

B. Triangular coordinate pattern

Following Balzarotti Eq. S24 [1], up to a global rotation and a relabelling of the exposures (centre last), the TCP has $K = 4$ exposures with zeros at

$$\mathbf{b}_k = \frac{L}{2} (\cos \varphi_k, \sin \varphi_k), \quad \varphi_k = \frac{\pi}{2} + \frac{2\pi k}{3}, \quad (2)$$

$$k = 0, 1, 2, \quad \mathbf{b}_3 = \mathbf{0},$$

so that L is the *diameter* of the circle of peripheral zeros [Fig. 1(c)]. Each exposure uses the same beam displaced to $\mathbf{b}_i$, $I_i(\mathbf{r}) = I(|\mathbf{r} - \mathbf{b}_i|)$.

C. Photon statistics, background and SBR

For a fixed total number N of detected photons the counts $(n_0, \dots, n_3)$ are multinomial with probabilities [1]

$$p_i^{(0)}(\mathbf{r}) = \frac{I_i(\mathbf{r})}{\sum_j I_j(\mathbf{r})}, \quad (3)$$

$$p_i(\mathbf{r}) = \frac{\text{SBR}}{\text{SBR} + 1} p_i^{(0)}(\mathbf{r}) + \frac{1}{\text{SBR} + 1} \frac{1}{K},$$

where the second form (Balzarotti Eq. S30) adds a background equal in all exposures. The SBR is the total signal over the total background summed over the K exposures at the emitter position (Eq. S29), so for a fixed

background it depends on the position and on the pattern. In all our simulations, however, we hold the SBR fixed at the quoted value at every position (Eq. S30 with constant SBR), which implies a position-dependent background; with a fixed background the SBR would instead vary with position (Eq. S29) and, at the centre, fall as L decreases (Eq. S32). When the beam carries a pedestal (finite zero depth, Sec. VIII) the SBR refers to the signal of the full beam, pedestal included. For the camera (Sec. VI) we use Balzarotti’s SBR_c , total signal over the background *per pixel*.

D. Fisher information and CRB

The Fisher information for the emitter position is

$$F(\mathbf{r}) = N \sum_{i=0}^{K-1} \frac{\nabla p_i \nabla p_i^T}{p_i}, \quad (4)$$

and we report the per-axis bound

$$\sigma_{\text{CRB}} = \sqrt{\frac{1}{2} \text{tr } F^{-1}}, \quad (5)$$

Balzarotti’s Eq. S13. The MC precision uses the same convention, $\sigma = [(\text{var } \hat{x} + \text{var } \hat{y})/2]^{1/2}$, and the root-mean-square error is $\text{rmse} = [(|\hat{\mathbf{r}} - \mathbf{r}|^2)/2]^{1/2}$. At the exact centre of the TCP without background the central exposure has $p_3 = 0$ and Eq. (4) is ill defined; Sec. IV distinguishes the *point value* at $r = 0$ from the *limit* $r \rightarrow 0$, and “centre CRB” means the limit unless stated otherwise.

E. Estimators

The MLE maximizes $\sum_i n_i \ln p_i(\mathbf{r})$ over a disk (grid search followed by a local refinement). The disk radius is part of the protocol and is stated with each result: L for centre points, $2L$ for positions swept across the pattern, and $0.75L$ for the iterative and misalignment studies. The LMS and modified LMS (mLMS) estimators are the linearized estimators of Balzarotti Eqs. S49–S51 [1], the latter with the polynomial coefficients (1.27, 3.8) used there for live tracking.

III. SCALAR VERSUS VECTORIAL DONUT (R1)

Zero and handedness. With circular polarization of the correct hand every field component of the focused vortex beam carries an azimuthal factor $e^{im\phi}$ with $m \neq 0$, so the on-axis intensity vanishes exactly at every z : the zero depth $I(0)/I_{\text{max}}$ is 0 to machine precision. With the opposite hand the longitudinal component E_z does not vanish on axis and the zero fills to 0.8452 of the ring maximum; with linear polarization the depth is 0.3716

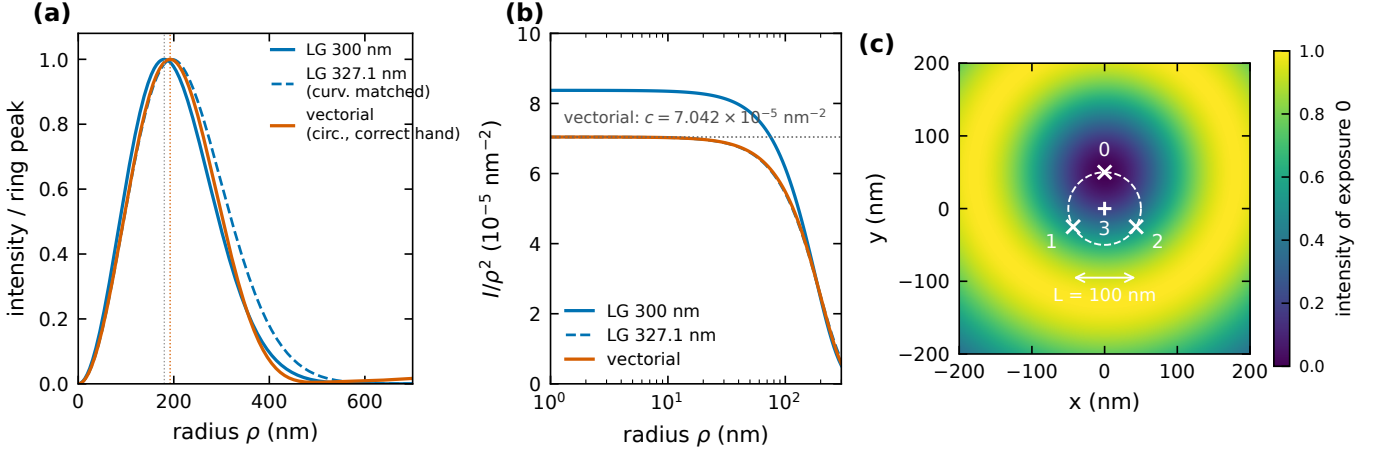


FIG. 1. Beam model and TCP geometry. (a) Normalized radial profiles of the scalar LG_{01} donut (fwhm = 300 nm, Eq. 1), of the LG donut with the zero curvature of the vectorial donut (fwhm = 327.1 nm), and of the vectorial Richards-Wolf vortex donut with circular polarization of the correct hand ($\lambda = 640 \text{ nm}$, $\text{NA} = 1.4$, $n = 1.518$, filling factor $5/3$); dotted lines mark the ring radii, i.e. half the peak-to-peak diameters 360.3 nm (LG) and 384.7 nm (vectorial). (b) I/ρ^2 near the zero on a logarithmic radius axis: all zeros are quadratic, the vectorial curvature is $c = 7.042 \times 10^{-5} \text{ nm}^{-2}$, and the curvature-matched LG follows the vectorial profile for $\rho \lesssim 100 \text{ nm}$. (c) TCP: three donut zeros (x) on a circle of diameter L (dashed; $L = 100 \text{ nm}$ shown for visibility) plus a central exposure (+); the map is the intensity of exposure 0.

[Fig. 2(a–d)], consistent with the experimental guidance on handedness of Refs. [7, 8].

Equivalent LG donut. The correct-hand vectorial ring has a peak-to-peak diameter 384.7 nm, and near the zero $I/I_{\text{max}} = c\rho^2$ with $c = 7.042 \times 10^{-5} \text{ nm}^{-2}$. The LG donut of Eq. (1) with the same curvature, $ea = c$, has fwhm = 327.1 nm; the LG donut with the same ring diameter would instead have fwhm = 320.3 nm.

CRB. Table I compares the centre CRB (limit $r \rightarrow 0$ defined in Sec. IV, $N = 100$, no background), with the vectorial beams evaluated exactly (no interpolation grid). The curvature-matched LG donut reproduces the vectorial CRB to the ratios listed in the last rows of the table, i.e. to a fraction of a percent up to $L = 150 \text{ nm}$; the default LG donut with fwhm = 300 nm overestimates it, with a ratio vectorial/LG of 0.9717 at $L = 150 \text{ nm}$ and a difference that keeps growing at larger L [Fig. 2(e,f)]. The overall curvature of the beam cancels in the probabilities p_i : any exactly quadratic profile gives the same CRB. The centre CRB is set by how the profile departs from quadratic at a distance $L/2$ from each zero [Eq. (A2), App. A]; with the peak normalized to one, the curvature-matched LG donut happens to reproduce this departure for the vectorial beam to a fraction of a percent. A filled zero, on the other hand, destroys the MINFLUX advantage, by a factor that depends on L (Table I): with the wrong hand the centre CRB rises from 1.600 to 147.9 nm at $L = 50 \text{ nm}$ and from 5.331 to 84.20 nm at $L = 150 \text{ nm}$, and with linear polarization to 60.77 and 35.72 nm, respectively; the relative loss is largest at small L .

IV. THE CRB AT THE TCP CENTRE

A. Point value and limit

Write $x \equiv L^2 \ln 2 / \text{fwhm}^2$ and $g \equiv 1 - x$ ($x < 1$, i.e. the peripheral zeros lie inside the ring maximum). At $\mathbf{r} = \mathbf{0}$ the central exposure has $p_3 = 0$ and $\nabla p_3 = 0$, so its term in Eq. (4) is 0/0. Dropping it gives the *point value*, Balzarotti’s Eq. S27 [1]:

$$\sigma_{\text{pt}} = \frac{L}{2\sqrt{2N}} g^{-1}. \quad (6)$$

For $\mathbf{r} \neq \mathbf{0}$, however, $p_3 \propto r^2$ and $\nabla p_3 \propto \mathbf{r}$, so the central term tends to a finite matrix of fixed norm along $\hat{\mathbf{r}}\hat{\mathbf{r}}^T$.

TABLE I. Centre CRB (nm; $r \rightarrow 0$ limit, $N = 100$, no background) of the TCP for the vectorial donut (correct hand, opposite hand, linear polarization) and for LG donuts with the default size parameter and with the curvature-matched one; last rows: CRB ratios.

L (nm)	50	100	150
vectorial, correct	1.600	3.323	5.331
LG, curv.-matched	1.601	3.330	5.351
LG, fwhm = 300 nm	1.605	3.363	5.486
vectorial, opposite	147.9	89.59	84.20
vectorial, linear	60.77	38.06	35.72
ratio corr./curv.	0.9995	0.9981	0.9962
ratio corr./LG default	0.9971	0.9881	0.9717

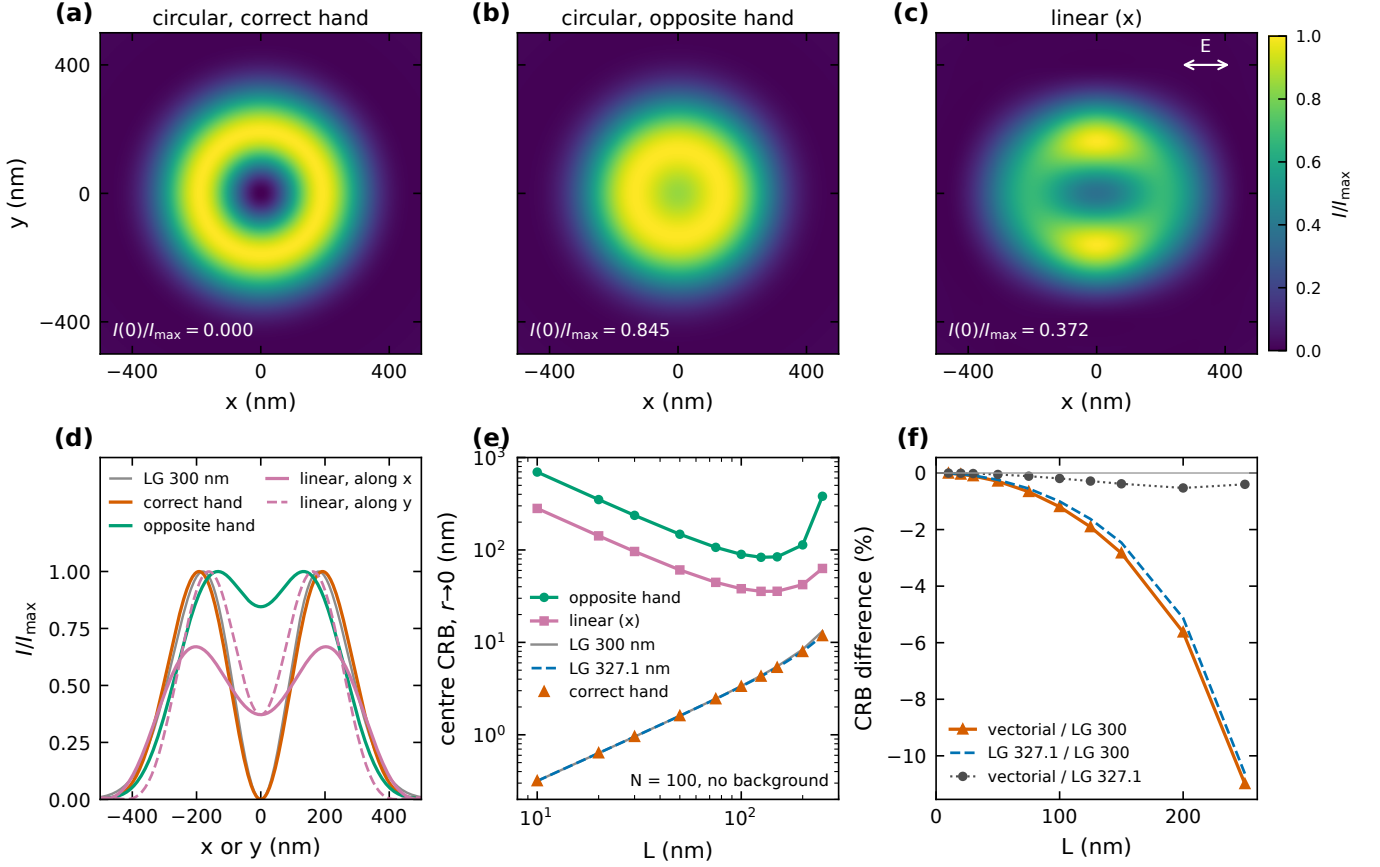


FIG. 2. Vectorial donut. (a–c) Focal-plane intensity, normalized to the ring maximum, for circular polarization of the correct hand, of the opposite hand, and for linear (x) polarization. (d) Profiles through the zero: the zero depth $I(0)/I_{\max}$ is 0 for the correct hand, 0.8452 for the opposite hand and 0.3716 for linear polarization. (e) Centre CRB ($r \rightarrow 0$ limit, $N = 100$, no background) of the TCP versus L ; all vectorial beams are evaluated exactly. (f) Relative CRB differences: the vectorial donut and the curvature-matched LG with respect to the LG donut with fwhm = 300 nm, and (grey) the vectorial donut with respect to the curvature-matched LG (fwhm = 327.1 nm), not to the LG with fwhm = 300 nm; the latter CRB ratio is 0.9962 at $L = 150$ nm.

The limit $r \rightarrow 0$ of the bound is (App. A)

$$\sigma_{\text{lim}}^2 = \frac{L^2}{8Ng^2} \frac{3g^2 + e^x}{3g^2 + 2e^x}, \quad \rho \equiv \frac{\sigma_{\text{lim}}}{\sigma_{\text{pt}}} = \sqrt{\frac{3g^2 + e^x}{3g^2 + 2e^x}}. \quad (7)$$

It does not depend on the direction of approach, so it is a well-defined, direction-independent “centre CRB” (the limiting error ellipse itself is anisotropic, see below). For $L = 50$ nm, $N = 100$ and fwhm = 300 nm the numerical limit of Eq. (5) (12 directions at $|\mathbf{r}| = 10^{-3}$ nm) is 1.605 nm and the closed form gives 1.605 nm, while the point value is 1.802 nm. The closed forms agree with an independent numerical Fisher computation in all tested cases. We use the limit because it describes an emitter at any small but non-zero distance from the centre; the point value describes a set of measure zero.

For a quadratic zero ($x \rightarrow 0$) the ratio is exactly $\rho = 2/\sqrt{5}$, i.e. $\sigma_{\text{lim}}^2 = L^2/(10N)$ against $\sigma_{\text{pt}}^2 = L^2/(8N)$: $\rho = 0.8944$ (numerically, with fwhm = 10^9 nm, 0.8944). With finite fwhm the ratio is not conserved: at fwhm = 300 nm

it is 0.8944, 0.8905, 0.8780 and 0.8553 for $L = 5, 50, 100$ and 150 nm [Fig. 3(f)], because the peripheral term scales as g^2 while the central one scales as e^x .

Origin of the discontinuity. The central term tends to $(\beta/N) \hat{\mathbf{r}} \hat{\mathbf{r}}^T$ (α and β are defined in Eq. (A4)), which has a limit in norm but not as a matrix; equivalently $\sqrt{p_3} \propto |\mathbf{r}|$ is a cone, so the differentiability of $\sqrt{p}$ required by the Cramér–Rao theorem fails exactly at $r = 0$. The central exposure contributes only radial information, of the same size at any distance (a quadratic zero is scale invariant), and a zero count is informative. The limiting error ellipse is therefore anisotropic: $\sigma_{\parallel} = (\alpha + \beta)^{-1/2}$ along $\hat{\mathbf{r}}$ and $\sigma_{\perp} = \sigma_{\text{pt}}$ across it (App. A); for $L = 50$ nm, $N = 100$ these are 1.380 nm and 1.802 nm; for a quadratic zero ($L = 100$ nm, $N = 100$) 2.739 nm and 3.536 nm, with ratio $\sqrt{3/5} = 0.7746$.

B. Background: continuity and non-commuting limits

With a finite SBR, $p_3 \geq (1-s)/K > 0$ with $s = \text{SBR}/(\text{SBR} + 1)$, the central term vanishes as r^2 and the bound is continuous; the point value equals the limit and Balzarotti's Eq. S31,

$$\sigma_{\text{CRB}}(\mathbf{0}) = \frac{L}{2\sqrt{2N}} g^{-1} \sqrt{\left(1 + \frac{1}{\text{SBR}}\right) \left(1 + \frac{3}{4\text{SBR}}\right)}. \quad (8)$$

For $\text{SBR} = 10$, $L = 50 \text{ nm}$ and $N = 100$, Eq. (8) gives 1.960 nm and the numerical limit 1.960 nm [Fig. 3(e)]. The two limits do not commute: $\lim_{\text{SBR} \rightarrow \infty} \lim_{r \rightarrow 0} \sigma_{\text{CRB}} = \sigma_{\text{pt}}$ whereas $\lim_{r \rightarrow 0} \lim_{\text{SBR} \rightarrow \infty} \sigma_{\text{CRB}} = \sigma_{\text{lim}}$; the central term switches on over a radius $r_c \propto \text{SBR}^{-1/2}$ that shrinks to a point [Fig. 4(c)]. To leading order $r_c \simeq (\sqrt{3}L/4) e^{-x/2}/\sqrt{\text{SBR}}$ (App. A): at realistic SBR it is not small compared with L (where the leading-order estimate is only indicative), and Eq. (8) is then the relevant centre value.

C. Maps and scaling

Figure 3(a–d) shows the CRB over the field of view for $L = 50$ and 100 nm , with and without background. The precision is best near the centre and degrades outside the circle of zeros.

Scaling with L and N . For $L \ll \text{fwhm}$ the centre CRB is linear in L : the log–log slope over $L = 5\text{--}40 \text{ nm}$ is 1.004 [Fig. 4(a)]. The exponent is range dependent: over larger L the factor g^{-1} steepens the curve, and with a perfect zero the bound diverges at the ring diameter $L = \text{fwhm}/\sqrt{\ln 2} = 360.3 \text{ nm}$ and decreases beyond. The dependence on N is exactly $N^{-1/2}$ (slope -0.5000 over $N = 10^2\text{--}10^4$) for the limit, Eq. (6) and Eq. (8) [Fig. 4(b)].

Photon cost of background. The photons needed for a 5 nm centre CRB at $L = 50 \text{ nm}$ (point value, Eqs. (6) and (8)) are 13.00, 13.45, 14.16, 15.37 and 17.93 for $\text{SBR} = \infty, 50, 20, 10$ and 5 ; with the background-free limit it would be 10.31 .

Comparison with published values and variants. With $N = 500$ and $\text{SBR} = 5$, Eq. (8) gives $0.9413, 1.962$ and 3.167 nm for $L = 50, 100, 150 \text{ nm}$ when $\text{fwhm} = 360 \text{ nm}$. These match, up to rounding in the last digit at $L = 150 \text{ nm}$, the MINFLUX centre-precision column of Supplementary Table 1 of Masullo *et al.* [3]; that table legend states no SBR or N for this column, so $\text{SBR} = 5$, $N = 500$ and this fwhm are our inference from the other columns. By contrast, with $\text{fwhm} = 300 \text{ nm}$ the same formula gives $0.9469, 2.012$ and 3.370 nm . The one-dimensional two-exposure bound $L/(4\sqrt{N})$ (Balzarotti Eq. S22c) is 1.250 nm for $L = 50 \text{ nm}$, $N = 100$. With a multiphoton exponent c ($\lambda_i \propto I_i^c$ [5]) the point value is σ_{pt}/c : $3.536, 1.768$ and 1.179 nm for $c = 1, 2, 3$

($L = 100 \text{ nm}$, $N = 100$, quadratic zero); for $c > 1$ the central term vanishes at the origin and there is no discontinuity.

V. ESTIMATORS VERSUS THE CRB (R3)

All MC results in this section use $L = 50 \text{ nm}$, $\text{fwhm} = 300 \text{ nm}$, seed 42 and 10^4 repetitions unless stated otherwise; standard errors (SE) of σ are nonparametric bootstrap estimates.

MLE with background: efficient at the centre. With $\text{SBR} = 10$ the model is regular at the centre ($p_3 > 0$). For $N = 100$ and an MLE search disk of radius L , the MLE precision is $\sigma = 1.943 \pm 0.010 \text{ nm}$ against the bound 1.960 nm of Eq. (8): an efficiency $\sigma/\sigma_{\text{CRB}} = 0.991 \pm 0.005$. This is the value we define as the MLE efficiency at the centre; the choice $\text{SBR} = 10$ (a regular, physical case) is a declared convention (Sec. X).

MLE without background: biased and superefficient. Without background $p_3 \propto r^2$, the model is not regular at the centre, and the MLE beats the CRB limit: $\sigma = 1.348 \pm 0.007 \text{ nm}$, i.e. $\sigma/\sigma_{\text{lim}} = 0.840 \pm 0.004$ and $\sigma/\sigma_{\text{pt}} = 0.748 \pm 0.004$ at $N = 100$. The ratio does not approach one as N grows [Fig. 5(c)], so this is not a small- N effect. The price is a bias, toward the centre close to it [for $x_0 \lesssim 8 \text{ nm}$, Fig. 5(d)]: at $\mathbf{r} = (2, 0) \text{ nm}$ the mean error along x is $-0.344 \pm 0.008 \text{ nm}$ (4×10^4 repetitions) [Fig. 5(d)]. The CRB bounds only unbiased estimators of a regular model, so there is no contradiction; it means that, without background, neither the point value nor the limit is a bound on what an MLE reports near the centre.

LMS. Without background the LMS estimate at the centre has exactly the covariance of the point value, $L^2/(8Ng^2)$ per axis: in MC, $\sigma/\sigma_{\text{pt}} = 1.000 \pm 0.005$ and $\sigma/\sigma_{\text{lim}} = 1.123 \pm 0.005$ ($= 1/\rho$), constant in N [Fig. 5(c)]. With background, the expectation of the LMS of Balzarotti Eq. S50 without the $1/s$ correction is exactly $s = \text{SBR}/(\text{SBR} + 1)$ times that of the background-free LMS at every position (the constant background term cancels because $\sum_k \mathbf{b}_k = \mathbf{0}$). In Fig. 5(a,b) the LMS and mLMS include the $1/s$ correction. Beyond $x_0 \approx 15 \text{ nm}$ they are strongly biased toward the centre [Fig. 5(a)], because the linearization compresses the estimate, and their σ falls below the CRB [Fig. 5(b)]; this reflects the compression, not a gain in precision. Closer to the centre this does not hold: for $x_0 \lesssim 12 \text{ nm}$ the mLMS has σ above the CRB, and for $x_0 \approx 5\text{--}15 \text{ nm}$ its bias is small and slightly outward.

MLE away from the centre: an outlier tail at low N . With $\text{SBR} = 10$ and $N = 100$ the MLE is not efficient everywhere inside the TCP. At $\mathbf{r} = (15, 0) \text{ nm}$, with a search disk of radius $2L$, $\sigma/\sigma_{\text{CRB}} = 1.51 \pm 0.03$, caused by a tail of outliers (the bump in Fig. 5(b) for $x_0 \approx 10\text{--}20 \text{ nm}$); at $N = 1000$ it is 1.004 ± 0.006 . The size of the tail depends on the search radius, which must therefore be reported with any such number; the convergence of

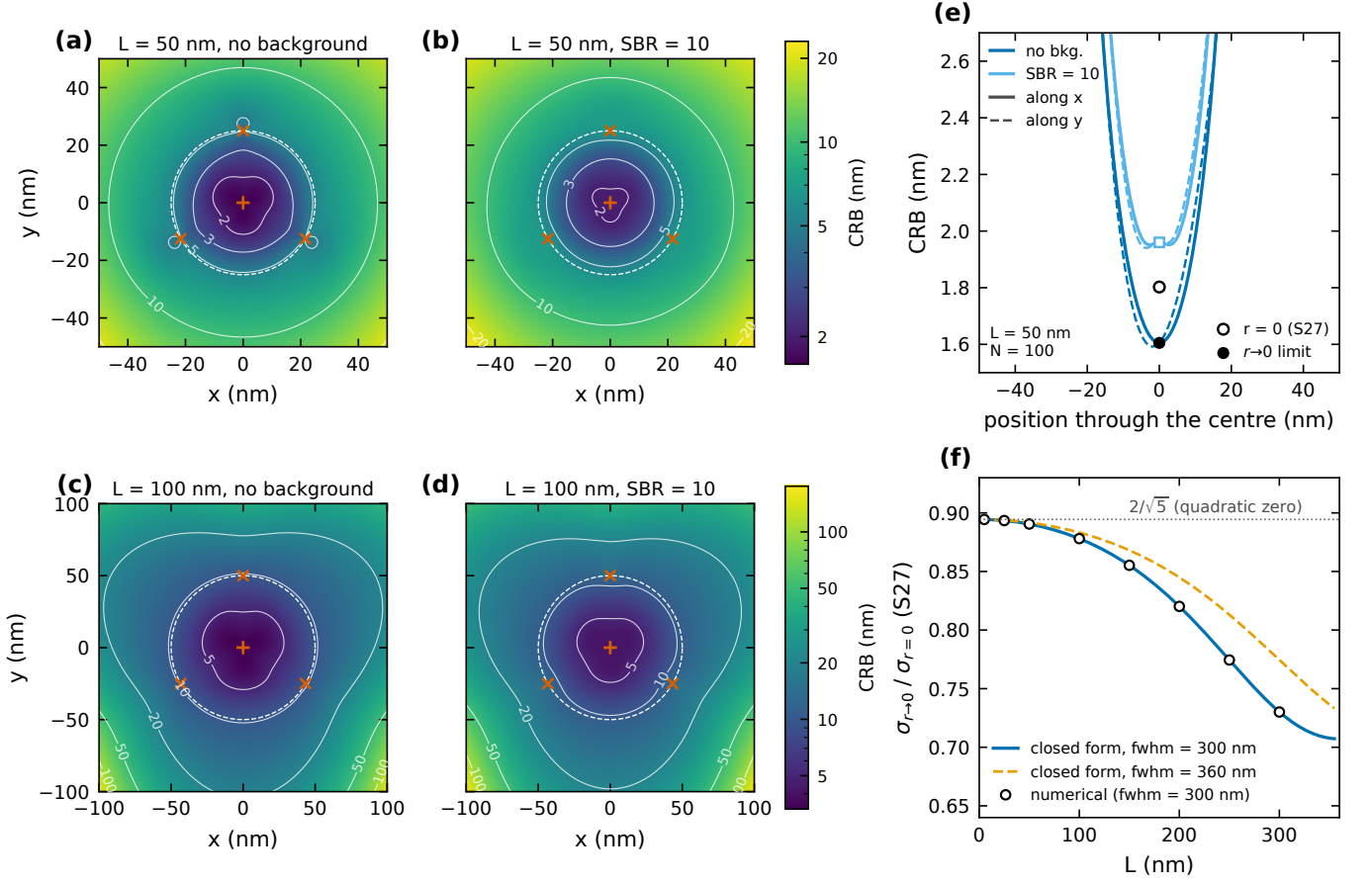


FIG. 3. CRB maps and the discontinuity at the TCP centre (fwhm = 300 nm, $N = 100$). (a–d) CRB over the field of view $|x|, |y| \leq L$ for $L = 50$ and 100 nm, without background and with SBR = 10 (logarithmic colour scale, contours in nm; pixels on a perfect zero take the $r \rightarrow 0$ limit). (e) Cuts through the centre along x and y for $L = 50$ nm: without background the $r \rightarrow 0$ limit (1.605 nm, filled circle) lies below the point value of Eq. (6) (1.802 nm, open circle), which excludes the central exposure with $p = 0$; with SBR = 10 the CRB is continuous (Eq. (8), 1.960 nm). (f) Ratio of the limit to the point value versus L , closed form (lines) and numerical Fisher computation (circles), with the closed form for fwhm = 360 nm dashed; it tends to $2/\sqrt{5}$ only for a quadratic zero ($L \ll$ fwhm).

the MLE to the CRB away from the centre only at larger N agrees with Ref. [1].

VI. CAMERA REFERENCE (R2)

As a reference we use the CRB of a camera with a Gaussian point-spread function of $\sigma_{\text{PSF}} = 100$ nm [1]. In the continuum limit without background the bound is $\sigma_{\text{PSF}}/\sqrt{N}$: 5.000 nm for $N = 400$ and 3.162 nm for $N = 1000$; this *ideal* value is the camera reference of Sec. VII. Pixelation costs little: a 9×9 window of 100 nm pixels gives 5.205 nm at $N = 400$ and 3.292 nm at $N = 1000$.

The SBR convention matters for the camera. Balzarotti's SBR_c is the total signal over the background *per pixel* (Eq. S61), so $\text{SBR}_c = 500$ on 9×9 pixels corresponds to a total SBR of 6.173. With that convention the pixelated camera reaches 4.964 nm at $N = 600$ and needs 591.4 photons for 5 nm; if 500 were the total SBR it

would reach 4.268 nm and need 437.1 photons. The SBR values of MINFLUX (total/total) and of the camera (total/per pixel) are therefore not directly comparable. For comparison, MINFLUX with $L = 50$ nm and SBR = 10 needs 15.37 photons for a 5 nm centre CRB [Fig. 4(d)].

VII. ITERATIVE MINFLUX (R4)

Protocol. The emitter is placed uniformly in a disk of radius $L_0/4$ around the initial pattern centre. Four iterations use TCPs of geometrically decreasing diameter $L_k = 150 \rightarrow 25$ nm with an equal share of the total photon budget N_{tot} ; each pattern is re-centred on the previous estimate, and each estimate is an MLE in a disk of radius $0.75L_k$ around the current centre. Unless stated otherwise there is no background, and each point uses 10^4 repetitions (seed 42) with bootstrap SE. The iterative scheme was proposed qualitatively in Ref. [1] and

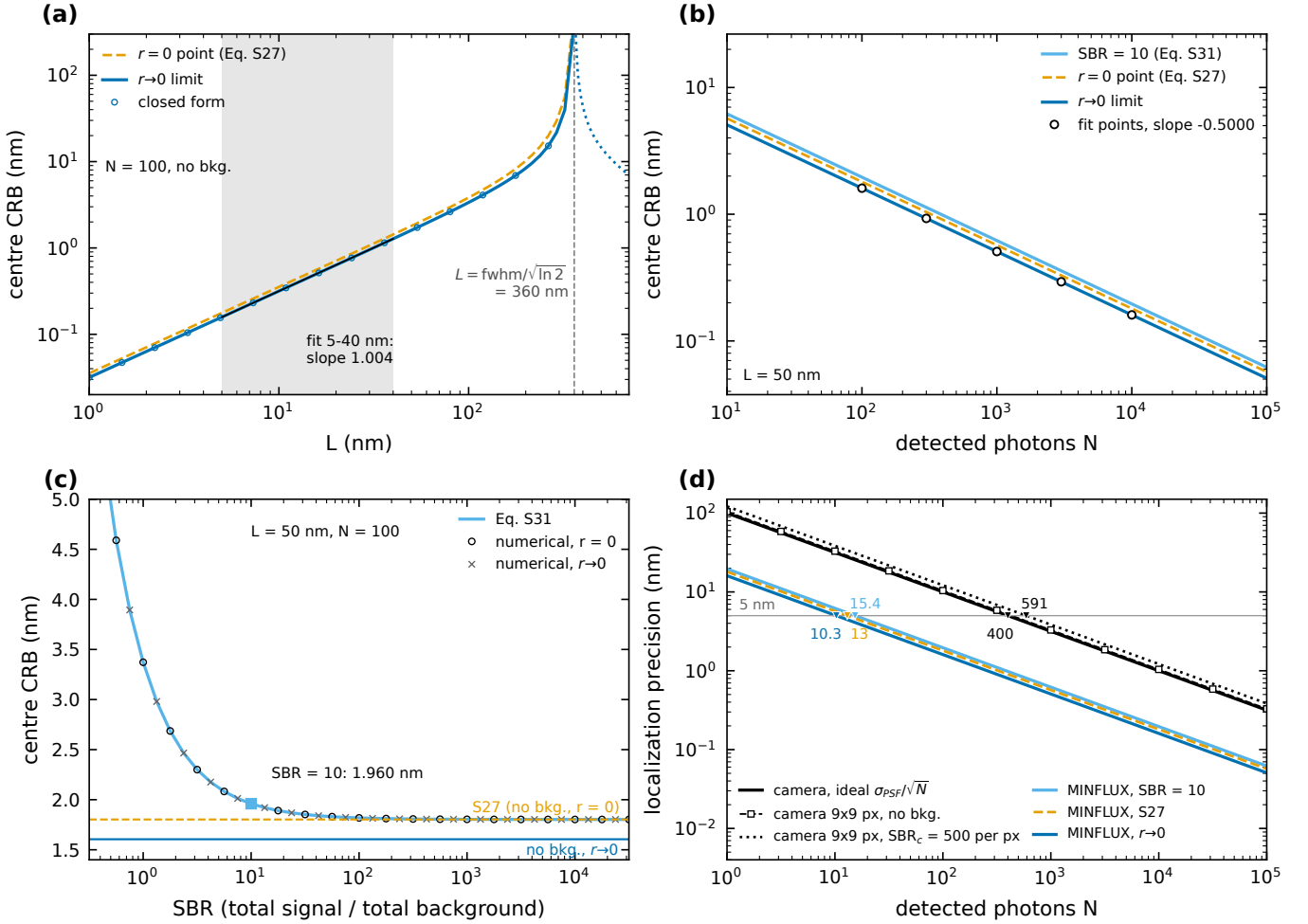


FIG. 4. Scaling of the centre CRB and comparison with camera localization. (a) CRB versus L ($N = 100$, no background): $r \rightarrow 0$ limit (numerical and closed form) and point value of Eq. (6); the log-log slope fitted over $L = 5-40$ nm (shaded, $L \ll \text{fwhm}$) is 1.004. With a perfect zero the CRB increases monotonically only for L below the ring diameter 360.3 nm, diverges there and decreases beyond (dotted). (b) CRB versus N for $L = 50$ nm: slope $-1/2$ for the limit, the point value and Eq. (8) with SBR = 10. (c) CRB versus SBR ($L = 50$ nm, $N = 100$): for finite SBR the point value equals the limit (Eq. (8)) and, as $\text{SBR} \rightarrow \infty$, tends to the point value of Eq. (6), not to the background-free limit. (d) MINFLUX ($L = 50$ nm) versus a camera ($\sigma_{\text{PSF}} = 100$ nm): continuum limit $\sigma_{\text{PSF}}/\sqrt{N}$ and a 9×9 window of 100 nm pixels, without background (dashed, nearly coincident with the continuum line) and with $\text{SBR}_c = 500$ defined per pixel (total SBR 6.173); triangles mark the photons needed for 5 nm: 400 for the ideal camera (5.000 nm at $N = 400$), 591.4 for the pixelated camera with background, and for MINFLUX 10.31 with the $r \rightarrow 0$ limit, 13.00 with the point value and 15.37 with SBR = 10.

relies on the scale invariance of a quadratic zero [6].

Precision and camera comparison. For $N_{\text{tot}} = 1000$ the final precision is $\sigma = 0.508 \pm 0.003$ nm, i.e. 0.1606 ± 0.0008 times the ideal camera value 3.162 nm with the same photons; with SBR = 10 it is 0.616 ± 0.003 nm. For reference, spending all photons at the final $L = 25$ nm with the emitter at the centre would give 0.2509 nm; the iterations pay for their field of view with photons spent at large L [Fig. 6(a,c)].

Photon scaling. Over $N_{\text{tot}} = 250, 500, 1000, 2000, 4000, 8000$ the final precision is 1.158, 0.736, 0.508, 0.3544, 0.2495 and 0.1763 nm. The log-log slope for $N_{\text{tot}} \geq 500$ is -0.515 ± 0.003 (bootstrap 95% interval $[-0.5196, -0.5096]$), which excludes $-1/2$: the iterative

scheme improves slightly faster than $N^{-1/2}$ (we have not isolated the mechanism; a plausible one is the better centring of the later iterations). Over the full range the slope is -0.537 ($[-0.5424, -0.5312]$), but it is driven by a heavy-tailed error distribution at $N_{\text{tot}} = 250$, where the bootstrap SE (0.014 nm) is itself unstable. The ratio to the ideal camera falls from 0.1831 at $N_{\text{tot}} = 250$ to 0.1577 and flattens [Fig. 6(b)].

Adaptive schedule. A rule $L_{k+1} = \max(L_{\min}, \kappa \sigma_k)$ with $\kappa = 6$ and σ_k the centre CRB of iteration k gives $L = 150, 25, 25, 25$ nm and $\sigma = 0.480 \pm 0.003$ nm at $N_{\text{tot}} = 1000$. The rule is less safe than it looks: the centre CRB of iteration 0 (3.470 nm) underestimates the real error (4.26 ± 0.03 nm), because the emitter is not at the

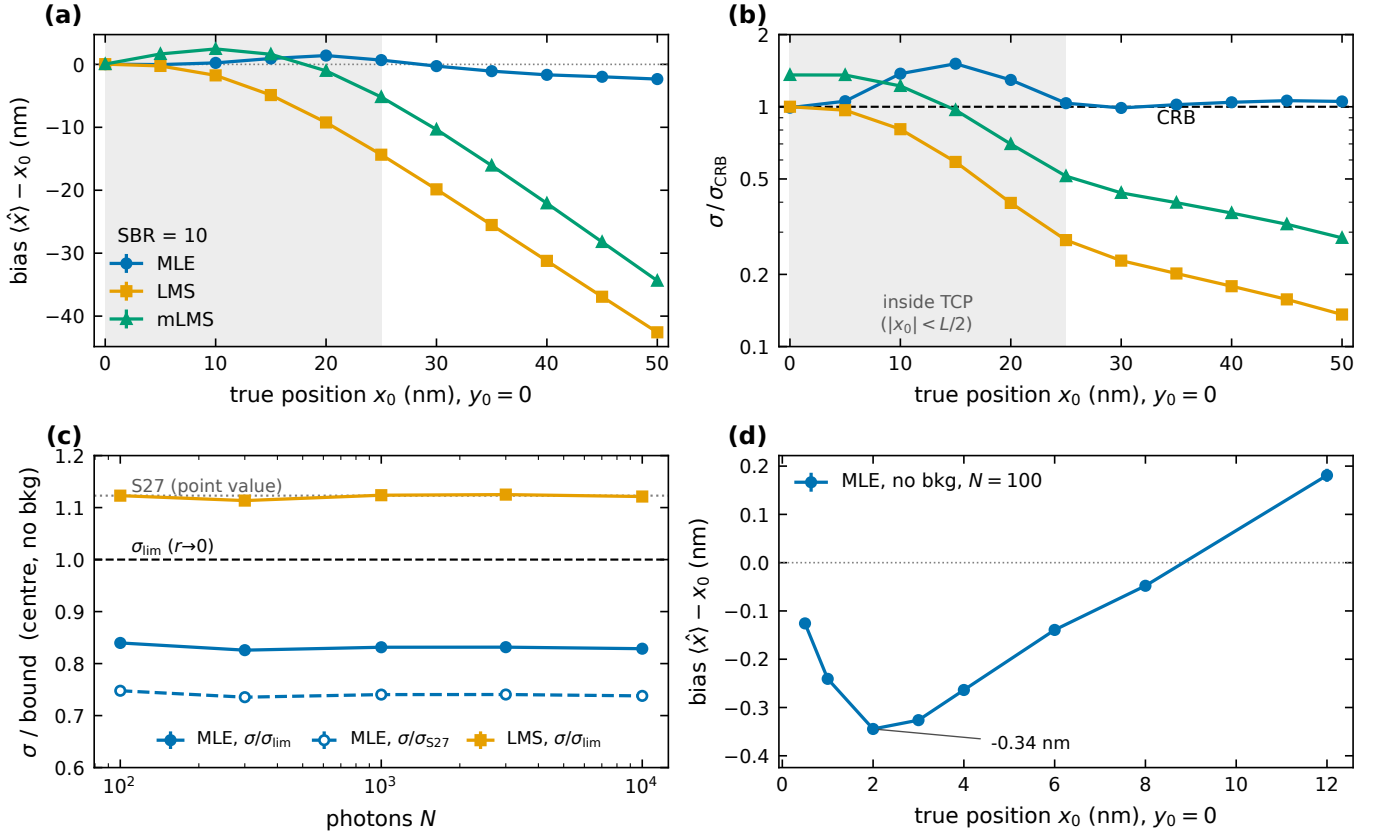


FIG. 5. Estimators versus the CRB ($L = 50$ nm, $N = 100$ unless indicated, seed 42). (a) Bias and (b) $\sigma/\sigma_{\text{CRB}}$ versus the emitter position x_0 along x , from the centre to L (inside the TCP shaded), for the MLE, the LMS and the mLMS with $\text{SBR} = 10$; off-centre MLE points use a search disk of radius $2L$, and the centre point (radius L , 10^4 repetitions) equals the efficiency 0.991. The MLE bump in (b) at $x_0 \approx 10$ – 20 nm ($\sigma/\sigma_{\text{CRB}} = 1.51$ at $x_0 = 15$ nm) is an outlier tail whose size depends on the $2L$ search radius. (c) Centre, no background: σ of the MLE divided by the $r \rightarrow 0$ limit (superefficiency, 0.840 at $N = 100$) and by the point value of Eq. (6) (0.748), and σ of the LMS divided by the limit, versus N . (d) Bias of the background-free MLE near the centre versus x_0 ($N = 100$): negative (toward the centre) for $x_0 \lesssim 8$ nm, with -0.344 ± 0.008 nm at $\mathbf{r} = (2, 0)$ nm, and changing sign (away from the centre) near $x_0 \approx 9$ nm.

centre of the first pattern, and a fraction 0.0265 ± 0.0016 of the emitters ends up outside the radius $L_1/2$ of the next pattern.

Without re-centring. If the patterns are not re-centred the final σ is 8.14 nm and worsens in the last iteration. This is an artefact of the protocol, not physics: the MLE search disk ($0.75L_k$) becomes smaller than the spread of emitter positions ($L_0/4$), and the estimates are truncated. The point is shown in grey in Fig. 6(a) and not used elsewhere.

VIII. NON-IDEALITIES (R5)

A. Finite zero depth

Experimental donuts have a residual intensity ϵ at the zero, relative to the ring maximum (below 0.2% in Ref. [1]). We model it in two ways (App. A): a *constant* pedestal, $I = I_{\text{LG}} + \epsilon$, and a *Gaussian* pedestal,

$I = (eap^2 + \epsilon)e^{-a\rho^2}$. The constant pedestal is exactly a background: at the centre the CRB is Eq. (8) with $\text{SBR}_\epsilon = 3ex e^{-x}/(4\epsilon)$; the Gaussian pedestal is not equivalent to any background, because it differs between exposures and reduces the slope of the donut. With $\epsilon > 0$ the point value equals the limit and tends, as $\epsilon \rightarrow 0$, to the point value σ_{pt} , not to σ_{lim} .

Degradation at fixed L . Table II lists $\sigma_{\text{CRB}}(\epsilon)/\sigma_{\text{pt}}$ at the centre. The loss is largest for small L , since $\text{SBR}_\epsilon \propto x \propto L^2$: at $L = 50$ nm a pedestal of 5% roughly doubles the CRB. The two pedestal models agree to first order in ϵ up to corrections of relative order x^2 .

Optimal L . Because shrinking L makes the pedestal relatively larger, the centre CRB (Gaussian pedestal, $N = 100$) has a minimum at an optimal diameter [Fig. 7(a)]: $L_{\text{opt}} = 10.48, 23.28, 50.34$ and 80.81 nm, with a CRB of 0.7458, 1.679, 3.878 and 7.287 nm, for $\epsilon = 0.002, 0.01, 0.05$ and 0.15 . In units of $\text{fwhm}\sqrt{\epsilon}$ the optimum is 0.7815, 0.7760, 0.7504 and 0.6955, independent of fwhm: the rule $L_{\text{opt}} \approx 0.7815 \text{ fwhm}\sqrt{\epsilon}$ holds only

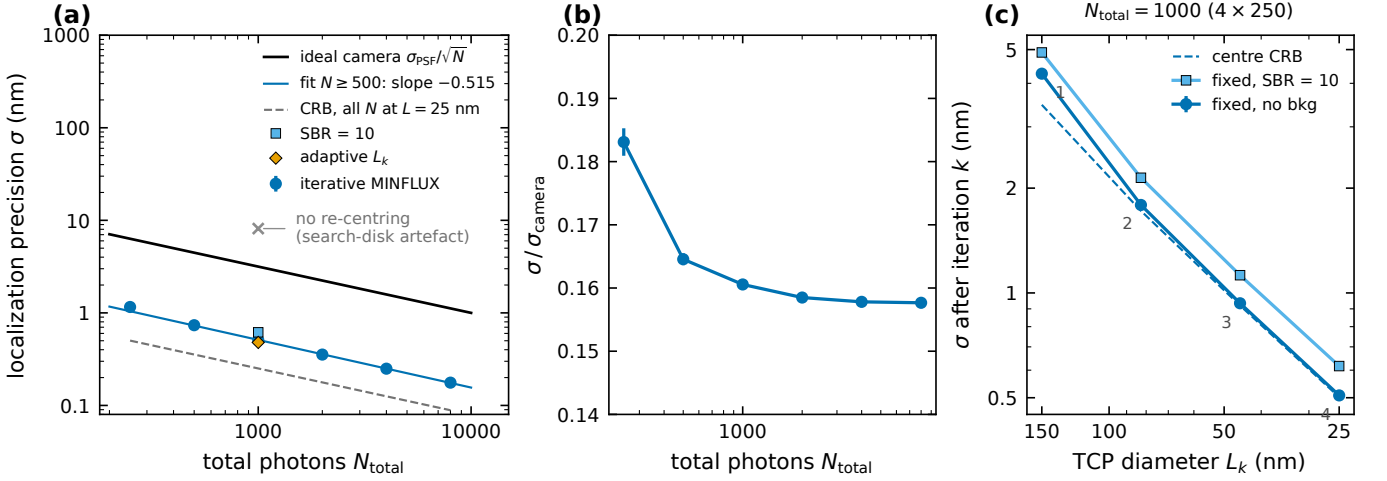


FIG. 6. Iterative MINFLUX versus camera localization (4 iterations, $L_k = 150 \rightarrow 25$ nm, equal photon split, re-centring, MLE, no background, 10^4 repetitions). (a) Final precision ($\sigma \pm$ bootstrap SE) versus total photon budget, with the ideal camera $\sigma_{\text{PSF}}/\sqrt{N}$, the fitted slope for $N_{\text{tot}} \geq 500$ (-0.515), the centre CRB if all photons were spent at $L = 25$ nm, and single points with SBR = 10 and with the adaptive schedule at $N_{\text{tot}} = 1000$; the grey cross (no re-centring) is an artefact of the truncated search disk. (b) $\sigma/\sigma_{\text{camera}}$ versus N_{tot} , from 0.1831 to 0.1577. (c) σ after each iteration versus the pattern diameter L_k for $N_{\text{tot}} = 1000$, without background and with SBR = 10, compared with the centre CRB of each iteration.

for $\epsilon \lesssim 0.01$ [Fig. 7(b)]. With $\epsilon = 0$ there is no optimum; the CRB grows monotonically up to the divergence at the ring diameter, 360.3 nm.

Off-centre minimum. With a nearly perfect zero the CRB minimum moves off the centre [Fig. 7(c), $\epsilon = 0.002$, $L = 100$ nm]: the pedestal dominates the zero within a radius of order $\sqrt{\epsilon/(ea)} = 4.887$ nm, independent of L . This is a scale, not the position of the minimum, which depends on the direction and on L .

B. Misalignment of the pattern

We displace each of the four zeros (including the central one) by δ in an independent random direction and localize with $L = 100$ nm, $N = 500$ and SBR = 10, using either the *naive* MLE, which assumes the nominal pattern, or the *honest* MLE, which uses the true one. MC statistics use 400 random patterns $\times$ 200 repetitions (seed 42), with standard errors computed between patterns, since the dispersion between patterns dominates.

TABLE II. Centre CRB with a finite zero depth ϵ , relative to the point value of Eq. (6) (fwhm = 300 nm, no background); constant / Gaussian pedestal.

ϵ	$L = 50$ nm	$L = 100$ nm
0.002	1.045 / 1.045	1.012 / 1.012
0.01	1.227 / 1.228	1.060 / 1.061
0.05	2.130 / 2.152	1.300 / 1.307
0.15	4.382 / 4.584	1.898 / 1.958

At $\delta = 0$ both estimators coincide exactly.

Bias. The naive estimator is biased roughly in proportion to δ [Fig. 8(a)]. At the centre the mean $|\text{bias}|$ is 1.49 ± 0.03 , 3.73 ± 0.08 and 7.71 ± 0.16 nm for $\delta = 2$, 5 and 10 nm, and at $(L/4, 0)$ 1.69 ± 0.04 , 4.20 ± 0.10 and 9.3 ± 0.3 nm. To separate the systematic bias from MC noise we also apply the naive MLE to the *expected* counts of each perturbed pattern (no Poisson noise), averaged over 4000 patterns. The resulting population slope of $|\text{bias}|$ versus δ (least squares through the origin over $\delta = 2, 5, 10$ nm) is 0.777 ± 0.005 at the centre and 0.851 ± 0.007 at $(L/4, 0)$ (SE from the sampling of patterns); both were reproduced by an independent computation with other seeds. The ratio $|\text{bias}|/\delta$ grows with δ , i.e. the bias is not exactly linear: at the centre it is 0.747, 0.755 and 0.783, and at $(L/4, 0)$ 0.802, 0.807 and 0.863 for $\delta = 2, 5, 10$ nm. As a complement, the MC slope at the centre (weighted least squares through the origin over the same δ values, noisy estimates) is 0.755 ± 0.009 ; it differs from the population estimate mainly through its inverse-variance weighting, which favours $\delta = 2$ nm, where $|\text{bias}|/\delta$ is smallest, and through the sampling of only 400 patterns, so the population value is the one we use.

The honest estimator removes the bias. At the centre its mean $|\text{bias}|$ at $\delta = 10$ nm, 0.180 ± 0.005 nm, is close to the MC noise floor $\sigma\sqrt{\pi/(2R)}$ expected from $R = 200$ repetitions alone, 0.1643 nm, but slightly above it, by about three SE. At $(L/4, 0)$ it is further above the floor (0.276 ± 0.008 nm against 0.2309 nm), so “honest at the floor” holds, approximately, only at the centre.

At the centre the loss is bias. The naive σ at the centre changes little, from 1.855 ± 0.003 nm at $\delta = 0$ to 2.015 ± 0.011 nm at $\delta = 10$ nm, while its rmse grows

to 6.228 nm [Fig. 8(b,c)]. At $(L/4, 0)$ the naive σ also grows, from 2.389 nm at $\delta = 0$ to 3.74 ± 0.16 nm at $\delta = 10$ nm, with a large dispersion between patterns, and the rmse reaches 8.978 nm. The honest σ follows the CRB of the true pattern at the centre (1.854 ± 0.009 nm against 1.842 nm at $\delta = 10$ nm) and lies slightly above it at $(L/4, 0)$ (2.606 ± 0.019 nm against 2.545 nm).

IX. DISCUSSION: WHAT MATTERS EXPERIMENTALLY

Ordering the effects studied by their numerical impact on the centre precision gives a clear hierarchy.

1. *The quality of the zero.* A filled zero is the largest effect by far. With the wrong handedness the centre CRB at $L = 50$ nm rises from 1.600 nm to 147.9 nm, and with linear polarization to 60.77 nm (Table I). Even a small residual ϵ matters at small L : at $L = 50$ nm a Gaussian pedestal of $\epsilon = 0.05$ multiplies the CRB by 2.152 (Table II). The practical consequence is that L should not be reduced below L_{opt} for the measured zero depth (Sec. VIII); polarization control of the donut [7] comes before any other refinement.

2. *Misalignment with a nominal model.* A displacement δ of the zeros that is ignored by the estimator produces a mean bias of $0.747\text{--}0.783\delta$ at the centre ($\delta = 2\text{--}10$ nm, $L = 100$ nm) and larger at $(L/4, 0)$ ($0.802\text{--}0.863\delta$). At $\delta = 10$ nm and $N = 500$ the bias at the centre (7.71 ± 0.16 nm) is several times the statistical precision (2.015 nm), and it is invisible in the spread of repeated localizations. Calibrating the actual pattern and using it in the likelihood removes it (honest MLE).

3. *Background.* At $L = 50$ nm, $N = 100$, SBR = 10 raises the centre CRB from 1.802 nm (point value) to 1.960 nm, and the photons for 5 nm from 13.00 to 15.37. Background also regularizes the model, making the MLE efficient at the centre.

4. *Estimator and protocol.* The MLE is efficient at the centre with background, but not uniformly inside the pattern at $N = 100$ ($\sigma/\sigma_{\text{CRB}} = 1.51$ at 15 nm from the centre), and the linearized LMS estimators are strongly biased off-centre. In iterative MINFLUX the search disk and the re-centring are part of the result; a heuristic L schedule based on the centre CRB leaves a fraction 0.0265 ± 0.0016 of emitters outside the next pattern.

5. *Definitions.* Without background the centre CRB depends on its definition by the factor $\rho = 0.8905$ at $L = 50$ nm; the background-free MLE even beats both values. Benchmarks against “the CRB at the centre” should state which one is used, or include background.

6. *Beam model.* Once the zero is correct, the scalar LG model is adequate if its size parameter is chosen to match the curvature of the actual zero: the error in the centre CRB is then below the percent level (Table I). With the default fwhm the discrepancy grows with L , reaching a CRB ratio of 0.9717 at $L = 150$ nm.

Iterative MINFLUX reaches an error 0.1577–0.1831

times that of the ideal camera for the same photons in our idealized setting; the zero depth, the misalignment and (except for one run with SBR = 10, which gives 0.616 ± 0.003 nm at $N_{\text{tot}} = 1000$) the background were absent from that simulation, and they set how much of this advantage survives in an experiment.

X. LIMITATIONS AND OPEN POINTS

The following points are either not verified or rest on declared choices; they are listed here rather than stated as results.

- *Scope of the model.* All simulations are two-dimensional, with a fixed N (multinomial statistics), a background equal in all exposures with the SBR held fixed at every position (Sec. II), no dipole orientation effects, no aberrations beyond the zero depth and no drift. The camera reference is ideal (Sec. VI); real cameras are worse.
- *rmse normalization.* We use the per-axis convention of Eq. (5) for σ and rmse. Whether it coincides with the error definition of Masullo *et al.* [3] was not checked, because the main text of that work is not in our notes.
- *SBR = 10 for the MLE efficiency.* The efficiency at the centre is defined with SBR = 10; this was chosen by the study coordinator on behalf of the author because without background the model is not regular (Sec. V). It is a convention, pending explicit confirmation by the author.
- *Vectorial ring diameter versus PyFocus.* Our peak-to-peak diameters of the vectorial donut as a function of the filling factor ($n = 1.5$) agree to within a few percent with the values we transcribed from Fig. 10 of Ref. [8] at filling factors 0.5, 2 and uniform illumination, but differ clearly at filling factor 1. We do not claim agreement there; an independent comparison is pending.
- *Heavy tail at low photon numbers.* In iterative MINFLUX at $N_{\text{tot}} = 250$ the error distribution is heavy-tailed and the bootstrap SE of σ is itself unstable; the full-range slope inherits this.
- *Range dependence of the L exponent.* The exponent 1.004 applies only to $L \ll \text{fwhm}$; it increases when the fit range extends to larger L .
- *Three-dimensional donuts and axial misalignment* are outside the scope of this study.

XI. METHODS AND REPRODUCIBILITY

Code. All computations use our own Python package `donutloc` (beams, patterns, photon model, Fisher infor-

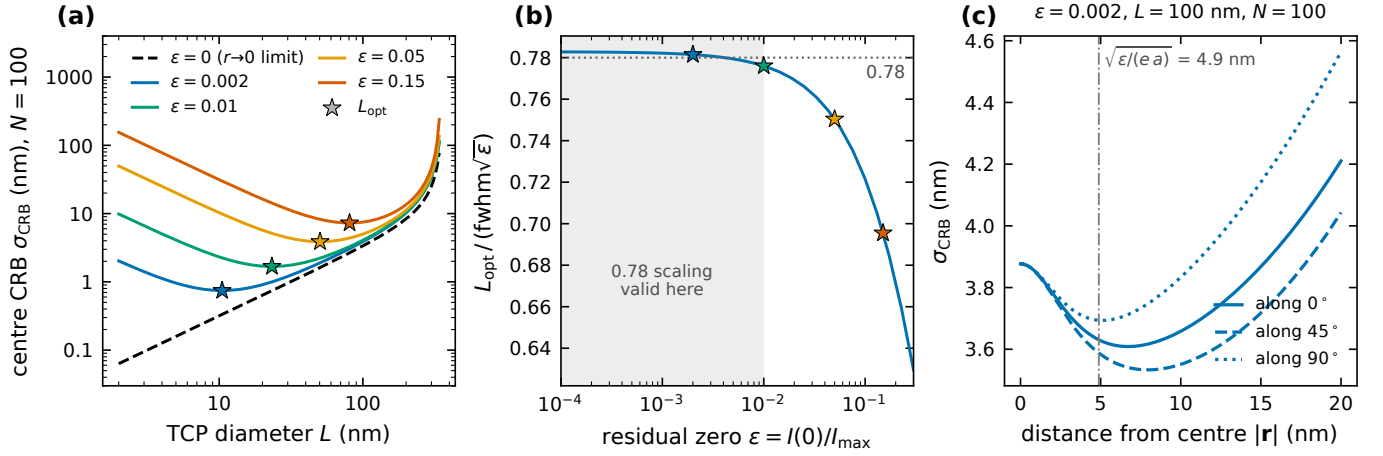


FIG. 7. Finite zero depth (fwhm = 300 nm, $N = 100$, Gaussian pedestal). (a) Centre CRB versus L for $\epsilon = 0.002, 0.01, 0.05, 0.15$ (stars: L_{opt}) and for $\epsilon = 0$ ($r \rightarrow 0$ limit), which has no optimum and diverges at the ring diameter. (b) $L_{\text{opt}}/(\text{fwhm}\sqrt{\epsilon})$ versus ϵ : the value 0.7815 at $\epsilon = 0.002$ holds only for $\epsilon \lesssim 0.01$ (shaded) and falls to 0.6955 at $\epsilon = 0.15$. (c) CRB (nm) versus distance from the centre along three directions for $\epsilon = 0.002$, $L = 100$ nm: the minimum lies off the centre; the dash-dotted line marks the transition scale $\sqrt{\epsilon/(ea)} = 4.887$ nm, a scale and not the position of the minimum.

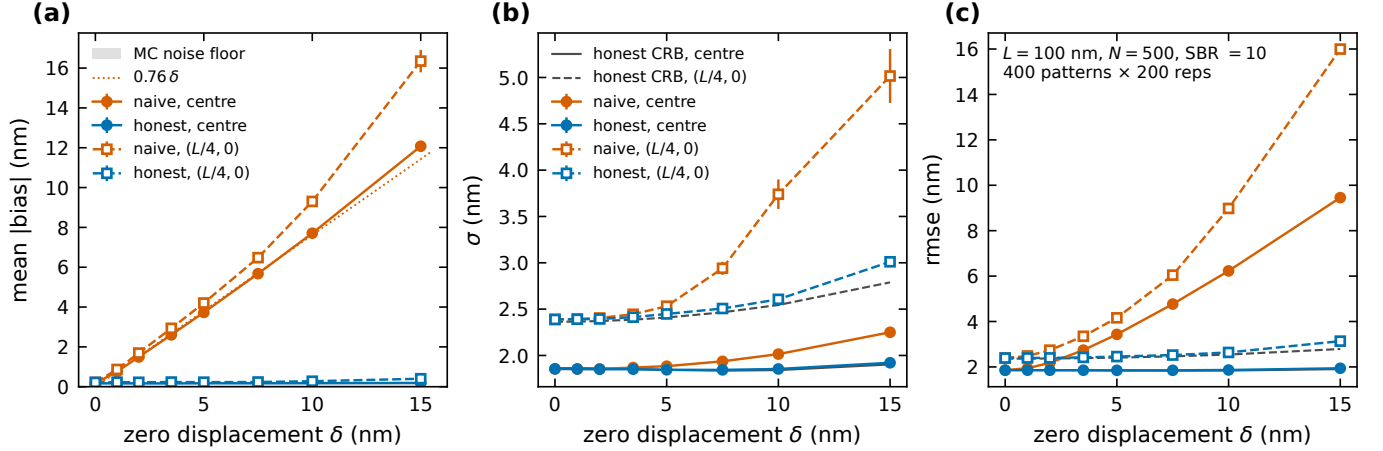


FIG. 8. TCP misalignment ($L = 100$ nm, $N = 500$, $\text{SBR} = 10$; all four zeros displaced by δ in random directions; 400 patterns $\times$ 200 repetitions; error bars are between-pattern SE). (a) Mean $|\text{bias}|$ of the naive MLE (nominal pattern) and of the honest MLE (true pattern) versus δ , at the centre and at $(L/4, 0)$, with the MC noise floor (grey band) and a straight line fitted to the naive centre points over all seven δ values including $\delta = 15$ nm (dotted). The main slope estimate is the noise-free population value, 0.777 at the centre and 0.851 at $(L/4, 0)$, growing with δ . (b) σ and (c) rmse versus δ , with the CRB of the true pattern: at the centre the naive σ changes little and its loss is bias, whereas at $(L/4, 0)$ the naive σ also grows, from 2.389 to 3.74 nm at $\delta = 10$ nm; the honest σ follows the CRB at the centre and lies slightly above it at $(L/4, 0)$.

imation, estimators, Monte Carlo, vectorial focusing, iterative MINFLUX), with NumPy and SciPy only. Each figure has its own script (`scripts/fig_*.py`); a single script, `scripts/compute_paper_numbers.py`, writes the registry `data/paper_numbers.json`, from which every number in this paper is typeset through generated macros; no computed result is typed by hand (only input parameters and exact constants are). All MC runs use the seed 42. The central results have a statistical error below 2%; the others are quoted with their SE, namely the misalignment quantities (where the dispersion between random patterns is intrinsic), the bias of the

background-free MLE at $\mathbf{r} = (2, 0)$ nm and the fraction of emitters left outside the next pattern by the adaptive L schedule. The closed forms of App. A are checked against a numerical Fisher computation by `scripts/verify_crb_closed_forms.py` and by unit tests. A reproduction script in `scripts/` runs, in order, the tests, the figures, the number registry and the provenance check.

Provenance. Every quantitative statement carries a tag that resolves in `paper/provenance.json` to the script, check or derivation that reproduces it; `scripts/check_provenance.py` verifies that every tag and every number resolves and that the generated macros are in

sync with the registry.

How the study was made. The study was carried out with an agent team following the agent-team methodology [9]: bounded rounds in which worker agents derive and compute, an independent verifier re-derives or re-computes every claim by a separate route before it is accepted, and a writer transcribes only verified claims with provenance-checked numbers. The complete audit trail (ledger of claims with their status, per-round reports and human decisions) is kept in `equipo/2026-09-26_donut-localization/` of the repository. The author set the questions, the conventions and the acceptance test.

ACKNOWLEDGMENTS

The simulations, derivations and the draft of this manuscript were produced with a team of AI agents under the agent-team methodology [9], with independent verification of every reported claim; the author directed the study and is responsible for its content.

Appendix A: Closed forms for the CRB at the TCP centre

This appendix transcribes the derivation in `docs/derivations/crb_tcp_center.md`; every boxed result there is implemented in `src/donutloc/closed_forms.py` and checked against a numerical Fisher matrix. Equation numbers S_n refer to the Supplementary Material of Ref. [1].

1. Notation

Write every exposure as $\lambda_i(\mathbf{r}) = h(|\mathbf{r} - \mathbf{b}_i|^2)$ with $\mathbf{b}_i$ from Eq. (2), $R = L/2$, and for the LG donut $h(u) = e a u e^{-au}$. Define $x \equiv aR^2 = L^2 \ln 2 / \text{fwhm}^2$ and $g \equiv 1 - x$. Since $\sum_k e^{i\varphi_k} = \sum_k e^{2i\varphi_k} = 0$,

$$\sum_{k=0}^2 \mathbf{b}_k = \mathbf{0}, \quad \sum_{k=0}^2 \mathbf{b}_k \mathbf{b}_k^T = \frac{3}{2} R^2 \mathbb{I}. \quad (\text{A1})$$

2. Master formula

With $h_R \equiv h(R^2)$, $h'_R \equiv h'(R^2)$ and $h_0 \equiv h(0)$, $\lambda_k = h_R - 2h'_R \mathbf{b}_k \cdot \mathbf{r} + O(r^2)$ for the peripheral exposures and $\lambda_3 = h_0 + h'(0)r^2 + O(r^4)$. By Eq. (A1) the linear term of $S = \sum_j \lambda_j$ cancels, $S = S_0 + O(r^2)$ with $S_0 = 3h_R + h_0$. At $\mathbf{r} = \mathbf{0}$, $\nabla p_k = -2h'_R \mathbf{b}_k / S_0$ and $\nabla p_3 = \mathbf{0}$, and the three peripheral terms of Eq. (4) give

$$F = \frac{6NR^2 h_R^2}{h_R(3h_R + h_0)} \mathbb{I}, \quad (\text{A2})$$

valid for any radial beam provided the central term vanishes at the origin. A background b per exposure amounts to $h_R \rightarrow h_R + b$ and $3h_R + h_0 \rightarrow S_0 + 4b$.

3. Point value

For the LG donut $h_R = exe^{-x}$, $R^2 h'_R = ex(1-x)e^{-x}$ and $h_0 = 0$. Equation (A2) gives $F = (8Ng^2/L^2) \mathbb{I}$, hence Eq. (6) (Eq. S27), with the central 0/0 term excluded.

4. Limit $r \rightarrow 0$

For $\mathbf{r} \neq \mathbf{0}$, $p_3 = ea r^2 / S_0 [1 + O(r^2)]$ and $\nabla p_3 = 2ea \mathbf{r} / S_0 [1 + O(r^2)]$, so the central term is

$$\frac{\nabla p_3 \nabla p_3^T}{p_3} = \frac{4ea}{S_0} \hat{\mathbf{r}} \hat{\mathbf{r}}^T + O(r^2) = \frac{16e^x}{3L^2} \hat{\mathbf{r}} \hat{\mathbf{r}}^T + O(r^2). \quad (\text{A3})$$

The $O(r)$ changes of the peripheral terms are odd in $\hat{\mathbf{r}}$ and do not contribute to the limit of the trace. Therefore

$$F_{\text{lim}}(\hat{\mathbf{r}}) = \alpha \mathbb{I} + \beta \hat{\mathbf{r}} \hat{\mathbf{r}}^T, \quad \alpha = \frac{8Ng^2}{L^2}, \quad \beta = \frac{16Ne^x}{3L^2}, \quad (\text{A4})$$

with eigenvalues $\alpha + \beta$ (along $\hat{\mathbf{r}}$) and α (across). The trace $\text{tr} F_{\text{lim}}^{-1} = 1/\alpha + 1/(\alpha + \beta)$ does not depend on $\hat{\mathbf{r}}$, and Eq. (5) gives Eq. (7). For $x \rightarrow 0$, $\beta/\alpha = 2/3$ and $\rho^2 = 4/5$, i.e. $\rho = 2/\sqrt{5}$ and $\sigma_{\text{lim}}^2 = L^2/(10N)$; for small x , $\rho \simeq (2/\sqrt{5})(1 - 9x/40)$. The axes of the limiting error ellipse are $\sigma_{\parallel} = (\alpha + \beta)^{-1/2}$ and $\sigma_{\perp} = \alpha^{-1/2} = \sigma_{\text{pt}}$, with ratio $\sqrt{3g^2/(3g^2 + 2e^x)}$, equal to $\sqrt{3/5}$ for a quadratic zero.

The discontinuity: the central term has norm β/N but direction $\hat{\mathbf{r}}$, so it has no matrix limit at $\mathbf{0}$ (its trace does); assigning it the value zero (the S27 convention) matches no directional limit. Equivalently, with $F = 4N \sum_i \nabla \sqrt{p_i} \nabla \sqrt{p_i}^T$, $\sqrt{p_3} \simeq \sqrt{ea/S_0} |\mathbf{r}|$ is a cone, not differentiable at the origin.

5. Background

Eq. S30 with $s = \text{SBR}/(\text{SBR} + 1)$ is identical to adding $b = S/(4\text{SBR})$ to each λ_i . With $h_0 = 0$, $h_R + b = h_R(1 + \frac{3}{4\text{SBR}})$ and $S_0 + 4b = S_0(1 + \frac{1}{\text{SBR}})$; Eq. (A2) gives Eq. (8) (Eq. S31). Now $p_3 \geq (1-s)/4 > 0$ and $\nabla p_3 = O(r)$, so the central term vanishes as r^2 : the CRB is continuous. Near the origin $\lambda_3 + b \simeq ea r^2 + b$ and the central term is $[4ea/(S_0 + 4b)] [r^2/(r^2 + r_c^2)] \hat{\mathbf{r}} \hat{\mathbf{r}}^T$ with $r_c = (\sqrt{3}L/4) e^{-x/2}/\sqrt{\text{SBR}}$ (leading order, $r_c \ll L$). Hence the non-commuting limits of Sec. IV.

6. Finite zero depth

Constant pedestal, $h = h_{LG} + \epsilon$: it is exactly a background $b = \epsilon$ for all $\mathbf{r}$, so at the centre

$$\sigma_{\text{CRB}}(\mathbf{0}) = \sigma_{\text{pt}} \sqrt{\left(1 + \frac{1}{\text{SBR}_\epsilon}\right) \left(1 + \frac{3}{4 \text{SBR}_\epsilon}\right)}, \quad (\text{A5})$$

$$\text{SBR}_\epsilon = \frac{3ex e^{-x}}{4\epsilon}.$$

With an additional background of given SBR measured against the full beam, pedestal included, $1 + 1/\text{SBR}_{\text{eff}} = (1 + 1/\text{SBR})(1 + 1/\text{SBR}_\epsilon)$ and the CRB is Eq. (8) at SBR_{eff} .

Gaussian pedestal, $h = (eau + \epsilon)e^{-au}$: $h_R = (ex + \epsilon)e^{-x}$, $R^2 h'_R = (exg - \epsilon x)e^{-x}$ and $h_0 = \epsilon$, so that

$$\frac{F_\epsilon}{F_{\text{pt}}} = \frac{3x^2(eg - \epsilon)^2}{g^2 D_\epsilon}, \quad D_\epsilon = (ex + \epsilon)[3(ex + \epsilon) + \epsilon e^x], \quad (\text{A6})$$

with $\sigma_{\text{CRB}} = \sigma_{\text{pt}} \sqrt{F_{\text{pt}}/F_\epsilon}$. It is not equivalent to a background: the pedestal is ϵ at the central exposure and ϵe^{-x} at the peripheral ones, and it reduces the slope by the factor $1 - \epsilon/(eg)$. To first order in ϵ the two models differ by a relative factor $1 + \frac{3}{7}x^2 + O(x^3)$ in the coefficient of ϵ . In both models $p_3(\mathbf{0}) > 0$, the CRB is continuous at the origin, and it tends to σ_{pt} as $\epsilon \rightarrow 0$. The transition radius $\sqrt{\epsilon/(ea)}$ is independent of L .

7. Multiphoton exponent

For $\lambda \propto I^c$, $h(u) = (ue^{-au})^c$, $h'_R/h_R = cg/R^2$ and $h_0 = 0$, so Eq. (A2) gives $F = 2Nc^2g^2/R^2$ and $\sigma_{\text{pt}}^{(c)} = Lg^{-1}/(2c\sqrt{2N})$. The central term scales as r^{2c-2} and vanishes at the origin for $c > 1$, so the limit equals the point value.

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